Stochastic Programming

Chapter I: Introduction. Examples of models

Jordi Castro
Group of Numerical Optimization and Modelling

Master in Statistics and Operations Research
Dept. of Statistics and Operations Research
Universitat Politècnica de Cataluna – Barcelona Tech

Slides based on book: J.R. Birge, F. Louveaux, *Introduction to Stochastic Programming*, Springer, 1997.

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1. The farming problem
   - General formulation of two-stage stochastic problems
2. The news vendor problem
3. Financial planning problem
4. Capacity expansion
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STOCHASTIC PROGRAMMING

A typical course in stochastic programming, following the book by Birge and Louveaux, may have the following sections:

- I - Modeling of stochastic programming problems $\rightarrow$ To be seen...
- II- Basic properties of stochastic programming problems $\rightarrow$ to be seen...
- III- Solution of SP problems $\rightarrow$ To be seen...
- IV- Additional aspects: other solution approaches (sampling methods, etc).

- PART I: STOCHASTIC MODELING
  - C.1 Introduction. Examples of models
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- PART II: BASIC PROPERTIES
  - C.3 Basic properties
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- PART III: SOLUTION OF SP PROBLEMS
  - C.5 Specialized algorithm: Benders decomposition (L-shapped method)

Chapter 1. INTRODUCTION. EXAMPLES OF MODELS

We will provide several case studies from the book of Birge and Louveaux.

- Different situations (deterministic, two stage, multistage etc.) of how an stochastic program can be solved .
- The value of the stochastic solution versus the deterministic solution will be analyzed.

The models to be seen are classified as:

(a) two-stage linear problems with recourse $\begin{cases} \text{(1) The farming problem} \\ \text{(2) The News vendor problem} \end{cases}$

(b) Multistage linear problems $\rightarrow$ (3) Simple problem of financial planning

(c) Chance or probabilistic constraints $\rightarrow$ (4) Capacity expansion of a power system

(d) Nonlinear S. P. $\rightarrow$ (5) Optimal design of a vehicle axle to ensure a certain quality

Deterministic problem

A farmer has 500 Ha of land where he could raising grain, corn and sugar beet. He knows that at least 200 tons (T) of wheat and 240 (T) of corn are needed for cattle feed. These amounts can be raised on the farm or bought from a wholesaler. Any production in excess of the feeding requirement would be sold. Selling prices are 170 € and 150 € per ton of wheat and corn respectively. Another profitable crop is sugar beet, which sells at 36 €/T; however, the European Commission imposes a quota on sugar beet production. Any amount in excess of the quota can be sold only at 10 €/T. The farmer's quota for next year is 6000T. Based on past experience, the farmer knows that the mean yield on his land is 2.5 T, 3 T, and 20 T per Ha for wheat, corn, and sugar beets, respectively. The next table summarizes these data and the planting costs for these crops.

	Wheat	Corn	Sugar beet
Yield (T/Ha)	2.5	3	20
Planting cost (€/Ha)	150	230	260
Selling price (€/T)	170	150	36 (< 6,000 T) 10 (> 6,000 T)
Purchase price (€/T)	238	210	-
Minimum requirement (T)	200	240	-
Total available land: 500 Ha			

The farmer wants to decide how much land to devote to each crop, to minimize costs.

Formulation

The linear problem has the following variables:

$x_1,\ x_2,\ x_3 =$ Ha of land devoted to wheat, corn, and sugar beets respectively.
$w_1,\ w_2 =$ tons of wheat, and corn to be sold (200 and 240 T).
$y_1,\ y_2 =$ tons of wheat, and corn to be purchased (200 and 240 T).
$w_3 =$ tons of sugar beets sold at the favorable price ($\leq$6,000 T).
$w_4 =$ tons of sugar beets sold at the lower price ($\geq$6,000T).

The problem reads as follows:

$$
\begin{array}{ll}
\min & 150x_1 + 230x_2 + 260x_3 + 238y_1 - 170w_1 + 210y_2 - 150w_2 - 36w_3 - 10w_4 \\
\text{s.to} & x_1 + x_2 + x_3 \leq 500 \ \rightarrow \ \text{[Total land]} \\
& 2.5x_1 + y_1 - w_1 \geq 200 \ \rightarrow \ \text{[Ensure wheat production]} \\
& 3x_2 + y_2 - w_2 \geq 240 \ \rightarrow \ \text{[Ensure corn production]} \\
& w_3 + w_4 \leq 20x_3 \ \rightarrow \ \text{[sugar beets production]} \\
& w_3 \leq 6,000 \ \rightarrow \ \text{[Production below 6,000 T]} \\
& x_1,\ x_2,\ x_3,\ y_1,\ y_2\ w_1,\ w_2,\ w_3,\ w_4 \geq 0
\end{array}
$$

Optimal solution based on expected yields

After solving this problem, the farmer obtains an optimal solution, as in the next table:

	Wheat	Corn	Sugar Beets
Surface (Ha)	$x_1 = 120$	$x_2 = 80$	$x_3 = 300$
Yield (T)	300	240	6,000
Sales (T)	$w_1 = 100$	$w_2 = 0$	$w_3 = 6000;\ w_4 = 0$
Purchase(T)	$y_1 = 0$	$y_2 = 0$	-
Overall profit: 118,600 €			

This optimal solution is easy to understand:

- The farmer devotes enough land to sugar beets to reach the quota of 6,000 T
- He then devotes enough land to wheat and corn production to meet the feeding requirement
- The rest of the land is devoted to wheat production. Some wheat can be sold

Modeling problems

The drawback of this model is that it does not take into account the variability of the land production whether one year is good, fair or bad. To consider this, we must formulate a stochastic approach. A model could be formulated differently depending on the assumptions we can make. We have two options:

- Consider the productivity of each of the three crops as discrete correlated random variables
- Consider the productivity of each of the three crops as continuous independent (uncorrelated) random variables (See Birge and Louveaux book, p. 11–15)

Stochastic modeling of the farmer's problem, considering discrete random variables

In this case we assume that the productivity of each crop is a random variable that takes three different values, all equally likely:

	Bad year	Normal year	Good year
r.v. productivity wheat →	2	2.5	3
r.v. productivity corn →	2.4	3	3.6
r.v. productivity sugar beets →	16	20	24

These three random variables are positively correlated (a good year for one, it is good for all). If the farmer knew if the coming year will be good, normal or bad, he would use the appropriate random variables values. What he can do is find out how sensitive is the solution of the problem to different types of years (or scenarios)

Stochastic modeling of the farmer's problem, considering discrete random variables

Solving the linear problem for the bad and the good years , we get:

Bad year

	Wheat	Corn	Sugar Beets
Surface (Ha)	$x_1 = 100$	$x_2 = 25$	$x_3 = 375$
Yield (T)	200	60	6,000
Sales (T)	$w_1 = 0$	$w_2 = 0$	$w_3 = 6,000$
			$w_4 = 0$
Purchase (T)	$y_1 = 0$	$y_2 = 180$	-
Objective function: -59,950 $\Rightarrow$ Overall profit: 59,950€			

Notice that any solution always exhaust sugar beets production to 6,000, but never exceeds the limit in which would sell at a lower price.

Good year

	Wheat	Corn	Sugar Beets
Surface (Ha)	$x_1 = 183.\widehat{3}$	$x_2 = 66.\widehat{6}$	$x_3 = 250$
Yield (T)	550	240	6,000
Sales (T)	$w_1 = 350$	$w_2 = 0$	$w_3 = 6,000$
			$w_4 = 0$
Purchase (T)	$y_1 = 0$	$y_2 = 0$	-
Objective function: -167,667 $\Rightarrow$ Overall profit: 167,667 €			

Stochastic modeling of the farmer's problem, considering discrete random variables

The optimal solution is very sensitive to changes in yields. The value of the variables $x_1,\ x_2,\ x_3$ varies greatly depending on the type of year or scenario. The problem is: What should we do if we do not know how it will be next year? To find the best solution for all the circumstances is the main target. That can be achieved as follows:

- The variables $x_1,\ x_2,\ x_3$ remain as before. These are decisions we make now, and they are called the *first stage decisions* (no stage replication)
- The variables w_i, $i = 1 \ldots 4$; y_j, $j = 1 \ldots 4$, once the decisions $(x_1,\ x_2,\ x_3)$ were taken, depend on the production obtained. Therefore they also depend on the values of the discrete random variables of productivity/scenario. To cover all cases, we created a new set of variables of the form $w_{is},\ i = 1 \ldots 4,\ \ s = 1 \ldots 3$ and $y_{is};\ i = 1,\ 2,\ \ s = 1 \ldots 3$ called the *second-stage variables*. $s = 1,\ 2,\ 3$ represents each particular stage ($s = 1$ bad year, $s = 2$ year s = normal, $s = 3$ good year)

Extensive form of the stochastic program

Assuming the farmer wants to maximize long-run profit, it is reasonable for him to seek a solution that maximizes his expected profit. If the three scenarios have an equal probability of $1/3$, the farmer's problem reads as follows:

$$
\begin{aligned}
\min \quad & 150x_1 + 230x_2 + 260x_3 - \tfrac{1}{3}(170w_{11} - 238y_{11} + 150w_{21} - 210y_{21} + 36w_{31} + 10w_{41}) \\
& \qquad - \tfrac{1}{3}(170w_{12} - 238y_{12} + 150w_{22} - 210y_{22} + 36w_{32} + 10w_{42}) \\
& \qquad - \tfrac{1}{3}(170w_{13} - 238y_{13} + 150w_{23} - 210y_{23} + 36w_{33} + 10w_{43}) \\
\text{s.to} \quad & \boxed{x_1 + x_2 + x_3 \leq 500} \quad \rightarrow \text{[Total land constraint]} \\
& 2x_1 + y_{11} - w_{11} \geq 200 \\
& 2.4x_2 + y_{21} - w_{21} \geq 240 \quad \rightarrow \text{[scenario 1 constraints]} \\
& w_{31} + w_{41} \leq 16x_3, \quad w_{31} \leq 6000 \\
& 2.5x_1 + y_{12} - w_{12} \geq 200 \\
& 3x_2 + y_{22} - w_{22} \geq 240 \quad \rightarrow \text{[scenario 2 constraints]} \\
& w_{32} + w_{42} \leq 20x_3, \quad w_{32} \leq 6000 \\
& 3x_1 + y_{13} - w_{13} \geq 200 \\
& 3.6x_2 + y_{23} - w_{23} \geq 240 \quad \rightarrow \text{[scenario 3 constraints]} \\
& w_{33} + w_{43} \leq 24x_3, \quad w_{33} \leq 6000 \\
& (x,\ y,\ w) > 0
\end{aligned}
$$

Optimal solution based on the stochastic model

The solution to the problem is as follows: (in red the solution using the expected value solution):

		Wheat		Corn		Sugar Beets	
Ha		$x_1 = 170$	120	$x_2 = 80$	80	$x_3 = 250$	300
$s = 3$ good year	Yield (T)	$3 \cdot 170 = 510$	360	$3.6 \cdot 80 = 288$	288	$24 \cdot 250 = 6000$	7200
	Sales (T)	$w_{13} = 310$	160	$w_{23} = 48$	48	$w_{33} = 6000$	6000
						$w_{43} = 0$	1200
	Purchase(T)	$y_{13} = 0$	0	$y_{23} = 0$	0		
$s = 2$ normal year	Yield (T)	$2.5 \cdot 170 = 425$	3000	$3 \cdot 80 = 240$	240	$20 \cdot 250 = 5000$	6000
	Sales (T)	$w_{12} = 225$	100	$w_{22} = 0$	0	$w_{32} = 5,000$	6000
						$w_{42} = 0$	0
	Purchase(T)	$y_{12} = 0$	0	$y_{22} = 0$	0		
$s = 1$ bad year	Yield (T)	$2 \cdot 170 = 340$	240	$2.4 \cdot 80 = 192$	192	$1.6 \cdot 250 = 4,000$	4800
	Sales (T)	$w_{11} = 140$	40	$w_{21} = 0$	0	$w_{31} = 4,000$	4800
						$w_{41} = 0$	
	Purchase(T)	$y_{11} = 0$	0	$y_{21} = 48$	48		0
Objective function: -108,389 $\rightarrow$ Optimal profit: 108,389 €					107,240 €		

Optimal solution based on the stochastic model

The optimal solution can be understood as follows:

- The area devoted to wheat, corn, and sugar beet, respectively is 170, 80 and 250.
- The most profitable decision for sugar beet land allocation is the one that always avoids sales at the unfavorable price even if this implies that some portion of quota is unused when yields are average or below average.
- The area devoted to corn is such that it meets the feeding requirement when yields are average.
- The area devoted to wheat is large enough to cover the minimum requirement.

This solution illustrates that it is impossible, under uncertainty, to find a solution that is ideal under all circumstances. The stochastic solution is not optimal for each scenario, but try to be good enough for all of them.

Solution comparison: Stochastic vs Deterministic

- Suppose that years variation is cyclical (bad year-normal year-good year). The farmer would then take optimal solutions and the expected benefit would be:

  $$\frac{\text{bad profits + normal profits + good profits}}{3} = \frac{59{,}950+118{,}600+167{,}667}{3} = 115,406$$

  The farmer unfortunately does not get prior information on the years. So the best he can do in the long run is to take the solution of the stochastic program. This leaves the farmer with an expected profit of €108,390

  The difference between this figure and the value €115,406, in the case of perfect information, namely $115,406 - 108,390 = 7,016$, represents what is called *the expected value of perfect information* (EVPI)
- Another approach the farmer may have is to assume expected yields and always to allocate the optimal planting surface according to these yields, as in the normal years: 2.5, 3 and 20 (T/Ha) for wheat, corn and sugar beets respectively. This approach represents the *expected value solution*. If the farmer use this solution every year results in a long run annual profit of 107,240 €. The loss by not considering the random variations is the difference between this and the stochastic model profit. This value, €108,390 - €107,240 = €1,150 is the *value of the stochastic solution* (VSS), the possible gain from solving the stochastic model.

General model formulation of a two-stage stochastic program with recourse

We may also use this example to illustrate the general formulation of a stochastic problem.

- We have a set of decisions to be taken without full information on some random events. These decisions are called *first-stage decision*. In the farmer example, they are the decisions on how many Ha to devote to each crop: $x_1,\ x_2,\ x_3$.
- Later, full information is received on the realization of some random vector ξ. In the farmer example ξ, represents the different yields linked to the different kind of year
- Then, second-stage or corrective actions y are taken. We use a functional form, such as $\xi(\omega)$ or $y(\omega)$, to show explicit dependence on an underlying event ω.

General model formulation of a two-stage stochastic program with recourse

In the farmer example, the random vector is the set of yields and the corrective actions, are purchases and sales of products. This defines the so-called two-stage stochastic program with recourse of the form:

$$\begin{array}{ll} \min & c'x + E_\xi Q(x,\xi) \\ \text{s.to} & Ax = b \\ & x \geq 0 \end{array} \quad \rightarrow \quad E_\xi Q(x,\xi) \text{ denote the mathematical expectation with respect to } \xi$$

Where:

$$\begin{array}{rll} Q(x,\xi) = \min & q^T(\omega)y(\omega) & \text{where } \xi = (q,h,T) \text{ (Random vector)} \\ \text{s.to} & Wy(\omega) = h(\omega) - T(\omega)x & \text{and } y \text{ is also a random vector} \\ & y(\omega) \geq 0 & \text{function of } \xi \end{array}$$

We assume here that W is fixed (*fixed recourse*)

General model of a two-stage St. Pr. with recourse

The farmer example can thus be written as:

$$\min\ [150\ 230\ 260] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + E_\xi Q(x,\xi)$$

$$\begin{array}{ll} \text{s.to} & x_1 + x_2 + x_3 \leq 500 \\ & x_i \geq 0 \end{array}$$

$$\begin{array}{rl} Q(x,\xi) = \min & 238y_1 - 170w_1 + 210y_2 - 150w_2 - 36w_3 - 10w_4 \\ \text{s.to} & t_1(\omega)x_1 + y_1 - w_1 \geq 200 \\ & t_2(\omega)x_2 + y_2 - w_2 \geq 240 \\ & w_3 + w_4 \leq t_3(\omega)x_3 \\ & w_3 \leq 6000 \\ & y_1,\ w_1,\ y_2,\ w_2,\ w_3,\ w_4 \geq 0 \end{array}$$

$$\text{where} \left\{ \begin{array}{lll} y & \rightarrow & \text{the second-stage decisions are } w_i \text{ and } y_i \\ T & \rightarrow & \text{associated to coefficients } x_i, \text{ in this case random} \\ W & \rightarrow & \text{associated to coefficients } y_i \text{ and } w_i, \text{ (constant} \rightarrow \text{ fixed recourse)} \\ h & \rightarrow & \text{associated to right hand side, in this case constant} \\ q & \rightarrow & \text{associated to the cost vector, in this case constant} \end{array} \right.$$

The random vector $\xi = (t_1(\omega),\ t_2(\omega),\ t_3(\omega))$ is formed by the three yields that ξ can take on three different values, say $\xi_1,\ \xi_2$ and ξ_3 (9 yields coefficients)

General model of a two-stage St. Pr. with recourse

The stochastic program can thus be written as:

- **Extensive form** given earlier for the farmer's example
- **Implicit representation** equivalent to the extensive form

$$\begin{array}{ll} \min & c'x + E_\xi Q(x,\xi) \\ \text{s.to} & Ax = b \\ & x \geq 0 \end{array}$$

- **Deterministic Equivalent form** to the implicit representation

$$\begin{array}{lll} \min & c'x + \mathcal{Q}(\text{x}) & \mathcal{Q}(\text{x}) \equiv E_\xi Q(x,\xi) \\ \text{s.to} & Ax = b & \\ & x \geq 0 & \end{array}$$

The news vendor problem

A news vendor goes to the publisher every morning and buys x newspapers at a price of c per paper. This number is bounded above by some limit u, representing either the news vendor's purchase power or a limit set by the publisher to each vendor. Problem description:

- Newspaper selling price is q ($q > c$).
- The unsold newspapers are returned to the publisher at the price of r ($r < c < q$).
- Demand for newspapers varies over days and is described by a random variable ξ.
- $y \rightarrow$ effective sales.
- $w \rightarrow$ number of newspaper returned to the publisher at the end of the day.

Formulation

We may formulate the problem as:

$$\begin{array}{ll} \min & cx + \mathcal{Q}(x) \\ \text{s.to} & 0 \leq x \leq u \end{array}$$

where $\mathcal{Q}(x) = E_\xi Q(x, \xi)$

and

$$\begin{array}{rl} Q(x,\xi) = \min & -qy(\xi) - rw(\xi) \equiv \max\ q \cdot y(\xi) + r \cdot w(\xi) \\ \text{s.to} & y(\xi) \leq \xi \\ & y(\xi) + w(\xi) \leq x \\ & y(\xi),\ w(\xi) \geq 0 \end{array}$$

The model illustrates the two-stage of the news vendor problem. The buying decision has to be taken before any information is given on the demand. When demand is known in the so-called second stage, which represents the end of the sales period of a given edition, the profit can be computed. This is done using the following rule:

$$\begin{array}{l} y^*(\xi) = \min(\xi, x) \\ w^*(\xi) = \max(x - \xi, 0) \end{array} \rightarrow \left\{ \begin{array}{lll} \text{if } x > \xi & \rightarrow & y^*(\xi) = \xi;\ \ w^*(\xi) = x - \xi \\ \text{if } x \leq \xi & \rightarrow & y^*(\xi) = x;\ \ w^*(\xi) = 0 \end{array} \right.$$

Formulation

$\mathcal{Q}(x)$ can be computed as:

$$\mathcal{Q}(\text{x}) = \int_{-\infty}^{\infty} g(\xi) f(\xi) d\xi \quad \text{where} \quad g(\xi) \left\{ \begin{array}{ll} \text{if } x > \xi & g(\xi) = -q\xi - r(x - \xi) \\ \text{if } x \leq \xi & g(\xi) = -qx \end{array} \right.$$

then:

$$\begin{array}{rl} \mathcal{Q}(\text{x}) & = \int_{-\infty}^{x} (-q\xi - r(x - \xi)) f(\xi) d\xi + \int_{x}^{\infty} (-qx) f(\xi) d\xi \\ & = (r - q) \int_{-\infty}^{x} \xi f(\xi) d\xi - rx \int_{-\infty}^{x} f(\xi) d\xi - qx \int_{x}^{\infty} f(\xi) d\xi \end{array}$$

Integrating by parts:

$$d(\xi F(\xi)) = F(\xi) d\xi + \xi F'(\xi) d\xi \ \rightarrow\ xF(x) = \int_{-\infty}^{x} F(\xi) d\xi + \int_{-\infty}^{x} \xi F'(\xi) d\xi$$

$$\rightarrow \boxed{\int_{-\infty}^{x} \xi F'(\xi) d\xi = xF(x) - \int_{-\infty}^{x} F(\xi) d\xi}$$

then:

$$\begin{array}{rl} \mathcal{Q}(\text{x}) & = (r - q) \left(xF(x) - \int_{-\infty}^{x} F(\xi) d\xi \right) - rxF(x) - qx(1 - F(x)) \\ & = -qxF(x) + rxF(x) + q \int_{-\infty}^{x} F(\xi) d\xi - r \int_{-\infty}^{x} F(\xi) d\xi - rxF(x) + qxF(x) - q(x) \\ & = \boxed{-qx + (q - r) \int_{-\infty}^{x} F(\xi) d\xi} \end{array}$$

where $\boxed{\mathcal{Q}'(x) = -q + (q - r)F(x)}$ and $\boxed{\mathcal{Q}''(x) = (q - r)f(x) \geq 0} \Rightarrow \mathcal{Q}(x)$ is convex

Solution

The deterministic equivalent is then the convex problem:

$$\begin{array}{ll} \min & (c-q)x + (q-r)\int_{-\infty}^{x} F(\xi)d\xi \\ \text{s.to} & 0 \le x \le u \end{array}$$

And the optimal solution is:

$$\begin{cases} x^* = 0 & \text{if } c - q + (q-r)F(0) > 0 \;\Leftrightarrow\; \frac{q-c}{q-r} < F(0) \\ x^* = u & \text{if } c - q + (q-r)F(u) < 0 \;\Leftrightarrow\; \frac{q-c}{q-r} > F(u) \\ x^* = F^{-1}(\frac{q-c}{q-r}) & \text{otherwise} \end{cases}$$

where $F^{-1}(\frac{q-c}{q-r})$ is the $\frac{q-c}{q-r}$-*quantile* of F

Example

Suppose $c = 8,\ q = 10,\ r = 5,\ u = 200$ and the demand is uniform on $[100,\ 200]$

$$\xi \sim U(100,\ 200) \;\Rightarrow\; F(\xi) = \begin{cases} 1 & \xi > 200 \\ \frac{\xi-100}{200-100} = \frac{\xi-100}{100} & \xi \in [100,\ 200] \\ 0 & \xi < 100 \end{cases}$$

$$\text{So:} \quad \begin{array}{l} F(0) = 0 \\ F(u = 200) = 1 \end{array} \qquad \frac{q-c}{q-r} = \frac{10-8}{10-5} = \frac{2}{5} = 0.4$$

Then $F(0) = 0 < 0.4 < 1 = F(u)$; $x^* = F^{-1}(0.4) = 140$
The value of the objective function is $x^* 140$

$$\begin{aligned} cx^* + \mathcal{Q}(x^*) &= 8 \cdot 140 + [-10 \cdot 140 + (10-5)\int_{-\infty}^{140} F(\xi)d\xi] \\ &= 8 \cdot 140 - 10 \cdot 140 + 5\int_{100}^{140} \frac{\xi-100}{100} d\xi \\ &= -2 \cdot 140 + \frac{5}{100}[\frac{\xi^2}{2} - 100\xi]|_{100}^{140} \\ &= -280 + \frac{5}{100}(\frac{140^2}{2} - 100 \cdot 140 - \frac{100^2}{2} + 100^2) \\ &= -280 + 5(-40 + \frac{140^2 - 100^2}{200}) = -240 \end{aligned}$$

$\Rightarrow$ The expected value of profit is $\boxed{240}$

VSS

What is the value of the stochastic solution in the example?

Considering "the expected value problem", we replace ξ by his expected value $E[\xi] \Rightarrow \xi = 150$ and constant.
Now, the problem is:

$$\begin{array}{ll} \min & cx - qy - rw = 8x - 10y - 5w \\ \text{s.to} & y \leq 150 \\ & y + w \leq x \\ & 0 \leq x \leq u \\ & y,\ w \geq 0 \end{array} \quad \Rightarrow \text{ The solution is: } \begin{array}{l} x^* = 150 \\ y^* = 150 \\ w = 0 \end{array}$$

To calculate VSS must apply the solution of the expected value problem, i.e. the value of the objective function of the stochastic formulation, and see what we get:

$$\begin{aligned} cx^* + \mathcal{Q}(x^*) &= 8 \cdot 150 - 10 \cdot 150 + \tfrac{5}{100} \int_{100}^{150} (0\xi - 100) d\xi \\ &= -2 \cdot 150 + \tfrac{5}{100} (\tfrac{150^2}{2} - 100 \cdot 150 - \tfrac{100^2}{2} + 100^2) \\ &= -300 + 5(-50 + \tfrac{150^2 - 100^2}{200}) = -237.5 \end{aligned}$$

$\Rightarrow$ The expected value of win is $\boxed{237.5}$

VSS $= 240 - 237.5 = 2.5$

EVPI

- $\forall \xi \rightarrow y^* = \xi, x^* = \xi, w = 0$
- $Cost(\xi) = 8\xi - 10\xi = -2\xi$
- $Cost = E_\xi[-2\xi] = -2E[\xi] = -2 \cdot 150 = -300.$
- $EVPI = -240 + 300 = 60$
- Naturally, we should have proper information about $E(\xi)$ to apply the last strategy. A good Statistician would be very helpful.

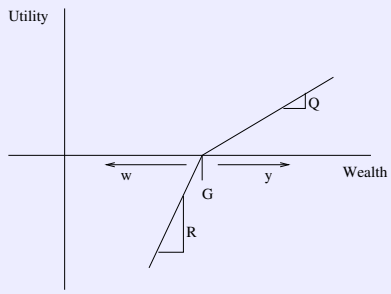
Financial planning and control

Financial decision-making problems can often be modeled as stochastic programs. In fact, the essence of financial planning is the incorporation of risk into investment decisions. This is one of the largest application areas of stochastic programming.

- We consider a simple financial planning problem where:
  - There are several stages, each of them with a certain random event and recourse variables
  - The random variables are in the constraint matrix instead of the right-hand side elements (as in the farmer example)
  - The decisions are highly dependent on past outcomes

Financial planning and control

- In the current problem suppose we want to invest an amount of money expecting at the end of the study period a certain amount.
  Data to consider:
  - B : Initial money to invest
  - G : Money that is expected at the end of the study period
  - T : Intervals of the time period. Money is reinvested at each stage interval.
  - $I = 2$: We have two types of investments (stocks or bonds)
- To determinate the objective function in mathematical terms we consider a utility function of money we have at the end of the period:



And the type of the objective function could be:

$$\boxed{\max \quad Qy - Rw}$$

y: excess variable (amount exceeding G)
w: slack variable (short for meeting the goal G)

Financial planning and control

- Suppose there are two possible investment products, stocks and bonds. The returns of each financial products is random and denoted by:

$$\xi(i,t) = \xi(i,t,\omega) \;\rightarrow\; \begin{array}{l} \text{return } i, \text{ interval } t \\ \omega \text{ is the random event} \end{array}$$

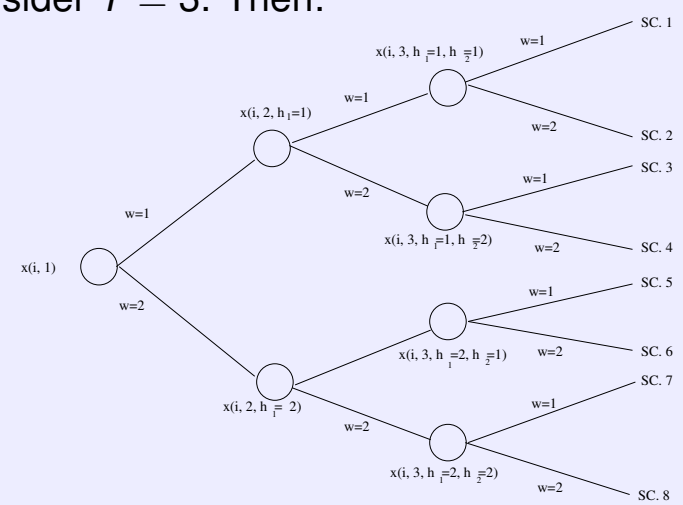
For simplicity we will assume that the random event can only take two values with equal likelihoods. This means that we have two types of return:

$$\text{When } \omega = 1 \;\rightarrow \left\{ \begin{array}{l} \xi(1,t) = 1.25 \;\rightarrow\; \text{stocks} \\ \xi(2,t) = 1.14 \;\rightarrow\; \text{bonds} \end{array} \right.$$

$$\text{When } \omega = 2 \;\rightarrow \left\{ \begin{array}{l} \xi(1,t) = 1.06 \;\rightarrow\; \text{stocks} \\ \xi(2,t) = 1.12 \;\rightarrow\; \text{bonds} \end{array} \right.$$

Financial planning and control

This would result in 2^T scenarios (2 values of ω, T time intervals). Consider $T = 3$. Then:



s=8 *scenarios*

We may refer to each scenario by the history of returns (ω):

$$\begin{array}{ll} So & Sc1 = (1,1,1) \\ & Sc2 = (1,1,2) \\ & \vdots \\ & Sc8 = (2,2,2) \end{array}$$

Note: unlike the farmer's problem, here we do not have only two stages , but T stages.

Financial planning and control

- The amount of money invested, are also random variables because they depend also on the history ω of earlier periods: $\boxed{x(i,t) = x(i,t,\omega)}$
  A key point about this investment model is that we cannot completely observe the random element ω when we make all our decisions $x(i,t,\omega)$. We can only observe the returns that have already taken place. This means that we cannot *anticipate* every possible outcome so our decisions are *nonanticipative* of future outcomes.
  To solve the problem we consider 2 alternatives in the definition of variables x:
  A.- Variables related to previous history: we link all the variables to past history, then:

$$\text{Decisions } x \rightarrow \begin{cases} x(i,1) \rightarrow \text{The amount invested at 1st period in product } i \\ x(i,2,h_1) \rightarrow \text{The amount invested at 2nd period in product } i \\ \qquad\qquad \text{knowing what happened in the period 1} \\ \vdots \\ x(i,T,h_1,\ldots,h_{T-1}) \rightarrow \text{The amount invested at } t=T \text{ period in} \\ \qquad\qquad \text{product } i \text{ knowing the past history} \\ \qquad\qquad \text{from } t=1 \text{ to } T=T-1 \end{cases}$$

$$\text{In general} \rightarrow \boxed{x(i,t,h_1,h_2,\ldots,h_{t-1})} \rightarrow \begin{cases} i: \text{ product} \\ t: \text{ period} \\ h_1,h_2,\ldots,h_{t-1}: \text{ past history} \end{cases}$$

Financial planning and control

$$\text{Slacks and surplus} \rightarrow \begin{cases} \boxed{y(h_1,h_2,\ldots,h_T)} \searrow \\ \boxed{w(h_1,h_2,\ldots,h_T)} \nearrow \end{cases} \text{Value based on the scenario, where we are}$$

$$\text{Probability scenarios} \rightarrow \left\{ \boxed{p(h_1,h_2,\ldots,h_T)} \right. \quad \text{Are known coefficients}$$

$$\text{Returns} \rightarrow \boxed{\xi(i,t,h_1,h_2,\ldots,h_t)} \rightarrow \begin{cases} i: \text{ product} \\ t: \text{ period} \\ \text{past history } (h_1,h_2,\ldots,h_{t-1}) \text{ and current}(h_t) \text{ in} \\ \text{ order to find its real value} \end{cases}$$

Note: The variables $x(i,1)$ would be the first stage. We made a decision and based on this and on

ω we are taking further decisions (2nd, 3th, ..., T-th stage)

Formulation

The formulation of the problem becomes:

$$\max \quad \sum_{h_1=1}^{2}\sum_{h_2=1}^{2}\cdots\sum_{h_T=1}^{2} p(h_1,h_2,\ldots,h_T)(Q\cdot y(h_1,h_2,\ldots,h_T)-R\cdot w(h_1,h_2,\ldots,h_T))$$

s.to [t=1]

$$\sum_{i=1}^{I} x(i,1)=B$$

[What we are reinvesting = what we have]

$$\sum_{i=1}^{I} x(i,t,h_1\ldots h_{t-1})=\sum_{i=1}^{I}\xi(i,t-1,h_1\ldots h_{t-1})x(i,t-1,h_1\ldots h_{t-2}) \quad \begin{cases}\forall h_1\ldots h_{t-1}\\ \forall t=2,\ldots T\end{cases}$$

[Relationship between y, w, money we have and G]

$$\sum_{i=1}^{I}\xi(i,T,h_1\ldots h_T)x(i,T,h_1\ldots h_{T-1})+w(h_1\ldots h_T)-y(h_1\ldots h_T)=G \quad \begin{cases}\forall h_1=1,\ 2\\ \forall h_2=1,\ 2\\ \vdots\\ \forall h_T=1,\ 2\end{cases}$$

$$x,\ y,\ w\geq 0$$

Financial planning and control

B.- Alternative version: variables related to stages

- Linking variables to the stage $s=1,\ldots,8$
- It is simpler than the previous one and allows a more comfortable modeling (with AMPL or any other system)
- We believe that the variables and coefficients depend on $s=1,\ldots,8$ and not past history. Then we have:

  Investment: $\rightarrow$ $\boxed{x(i,t,s)}$: money invested in i, in period t, under scenario s

  slack, surplus $\rightarrow$ $\begin{cases}\boxed{y(s)}\\ \boxed{w(s)}\end{cases}$

  Likely scenarios: $\rightarrow$ $\boxed{p(s)}$
  (coefficients)

  return (coefficients): $\rightarrow$ $\boxed{\xi(i,t,s)}$

- The difficulty with this formulation is that, when we have split up the scenarios, we may have lost nonanticipativity of the decisions because they would now include knowledge of the outcomes up to the end of the horizon. To enforce nonanticipativity, we add constraints explicitly in the formulation. For example:

Financial planning and control

$*$ $x(i,1,s)$ equal $\forall s = 1 \dots 8$

$*$ $x(i,2,s)$ equal $\begin{cases} \forall s = 1 \dots 4 \\ \forall s = 5 \dots 6 \end{cases}$

$*$ $x(i,3,s)$ equal $\forall \begin{cases} s = 1,2 \\ s = 3,4 \\ s = 5,6 \\ s = 7,8 \end{cases}$

So we can define:

$C_{s,t} = \{s' = (h'_1, \dots h'_T) |\ (h'_1, \dots h'_{t-1}) = (h_1, \dots h_{t-1}),\ \text{where } s = (h_1, \dots h_T)\}$

For example: $C_{2,3} = \{1,2\}$; $\quad C_{2,2} = \{1,2,3,4\}$; $\quad C_{2,1} = \{1,2,3,4,5,6,7,8\}$

Final formulation

- The final formulation becomes:

$$\max \quad \sum_{s=1}^{S} p(s)(Qy(s) - Rw(s))$$

$$\text{s.to} \quad \sum_{i=1}^{I} x(i,1,s) = B \ \ \forall s = 1 \dots S \qquad [t = 1]$$

$$\sum_{i=1}^{I} \xi(i,t-1,s)x(i,t-1,s) = \sum_{i=1}^{I} x(i,t,s) \qquad s = 1 \dots S, \ \ t = 2 \dots T$$

$$\sum_{i=1}^{I} \xi(i,T,s)x(i,T,s) + w(s) - y(s) = G \qquad \forall s = 1 \dots S$$

$$\sum_{s' \in C_{s,t}} p(s')x(i,t-1,s') = \left(\sum_{s' \in C_{s,t}} p(s') \right) x(i,t-1,s) \quad t = 1 \dots T$$

$$s = 1 \dots S$$
$$i = 1 \dots I$$

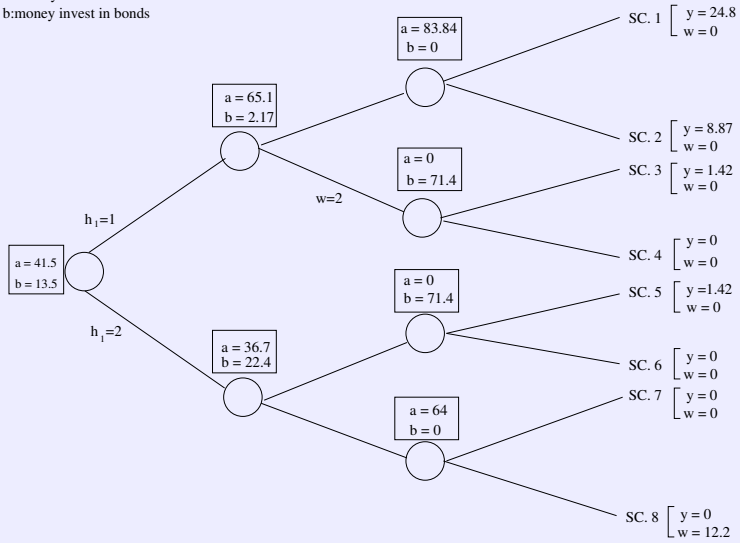
$$x(i,1,s) \geq 0 \quad \forall i,t,s$$
$$y(s) \geq 0,\ w(s) \geq 0,\ \forall s$$

Exercise

AMPL formulation in the two versions using the following data: $\begin{cases} T = 3 \\ B = 55 \quad (1000 \text{ of } €) \\ G = 80 \quad (1000 \text{ of } €) \\ Q = 1 \\ R = 4 \end{cases}$

And the the optimal solution is:

a: money invest in stocks
b:money invest in bonds



* The value of the objective function is: $\boxed{-1.52}$

Interpretation of the solution

- The initial solution is heavily in stock (41.5) with only 13.5 in bonds.
- These values change as we move in the scenario tree. In SC_1 to SC_4 stocks are more prominent than in SC_5 to SC_8.
- In the last period, the investments are either completely in stocks or completely in bonds. This occurs because in SC_1 and SC_2, there is no risk of missing the target. In SC_3 to SC_6, may cause one to miss the target, so they are avoided. In SC_7 and SC_8, the only hope of reaching the target is through stocks.

Expected value solution

- All random returns are replace by their expectations. Therefore the stock and bond values become:

$$\left.\begin{array}{l}\xi(1,t) = \frac{1}{2}\cdot 1.25 + \frac{1}{2}\cdot 1.06 = 1.155 \\ \xi(2,t) = \frac{1}{2}\cdot 1.14 + \frac{1}{2}\cdot 1.12 = 1.13\end{array}\right\} \textit{The optimal investment plan places all funds in stocks in each period}$$

- If we apply the solution of the expected value to the stochastic problem (setting $x(2,t,s) = 0\ \forall t, \forall s$). It is obtained a Obj.F=-3.78, while the stochastic program value=-1.52. The difference between this quantities is the value of the stochastic solution: VSS=-1.52-(-3.78)=**2.23**

Expected value solution Cont...

- Notice that the stochastic solution reaches the goal 7/8-100% = 87.5% of the time (we only get the scenario 8 below G). However, if we applied the expected value solution, in 50% of the time we do not reach the G target.

- Exercise: Compare with AMPL the results indicated before with the solution of the expected value (EV); to be done in the lab session.

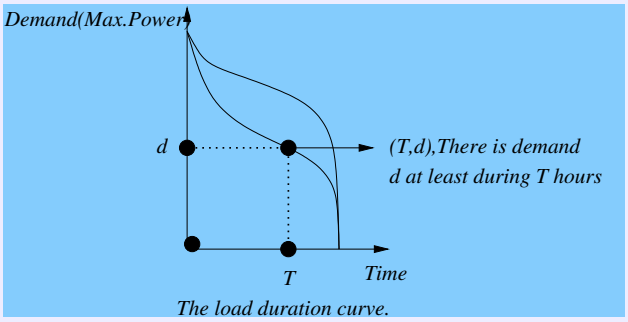
Capacity Expansion Model

- The capacity expansion problems consist in finding the way to invest and to satisfy a demand over the time in the most optimal way (minimum cost).
- Applications:
  - Power systems
  - Communication systems

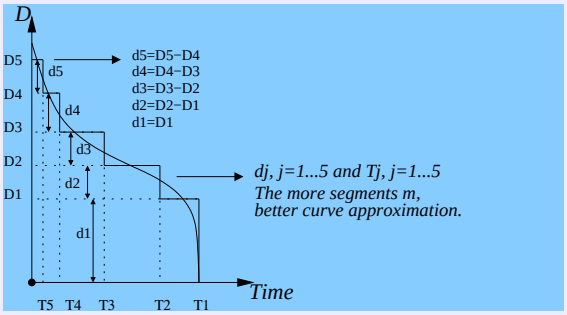
  ⋮
- Here it will be illustrated the case of power plant expansion for electricity generation. The following models will be studied:
  - 1st → Deterministic Static Model
  - 2nd → Stochastic Dynamic Model

Deterministic Model

- The demand (d) of an electric company over a period of time (T) is given by the load duration curve:



- This curve can be approximated by a staircase function:



Deterministic Model Cont...

- Supposing that we have n types of technologies to choose and in different modalities or segments m (nuclear, thermal, or eolic plants, etc...)
- Each segment j is defined by demand $\rightarrow d_j$ and duration $(time) \rightarrow \tau_j$
- Each type of technology has 3 associated parameters:

$$i = 1...n \begin{cases} r_i \rightarrow \textit{investment cost per Mw} \\ q_i \rightarrow \textit{operating cost per Mwh} \\ a_i \rightarrow \textit{availability factor of technology}(\textit{fail} = 0, \textit{available} = 1) \end{cases}$$

- Thus, in order to know which technology goes to each mode we can easily solve:

$$\forall j = 1 \ldots m \ \text{"find"} \quad i(j) = arg \min_{i=1\ldots n} \left\{ \frac{r_i \cdot d_j + q_i \cdot \tau_j \cdot d_j}{a_i} \right\} = arg \min_{i=1\ldots n} \left\{ \frac{r_i + q_i \cdot \tau_j}{a_i} \right\}$$

$$\textbf{Interpretation} \begin{cases} *\textit{for initial segments where } \tau_j \textit{ is large} \rightarrow \textit{use} \\ \quad \textit{technologies with small } q_i \\ *\textit{for final segments where } \tau_j \textit{ is small} \rightarrow \textit{use} \\ \quad \textit{technologies with small } r_i \end{cases}$$

Stochastic Model

Let's consider the following example:

- Parameters:
  - $T = 1$ (2 stage problem)
  - $n = 4$ (technologies)
  - $a = 1$ (availability)
  - $m = 3$ (three modalities or segments)
  - $d_{j=1\ldots m} = (\xi, 3, 2)$
  - Fixed cost: $r_{i=1\ldots n} = (10, 7, 16, 6)$
  - Operating cost: $q_{i=1\ldots n} = (4, 4.5, 3.2, 5.5)$
  - Time duration: $\tau_{j=1\ldots m} = (10, 6, 1)$
- Variables:
  - $x_i, i = 1...n$: investment in each technology i
  - $y_{ij}, i = 1...n,, j = 1...m$: part of demand j covered by technology i

Stochastic Model Cont...

$$
\begin{aligned}
\min \quad & \sum_{i=1}^{4} r_i \cdot x_i + E_\xi \Big[\sum_{j=1}^{3} \tau_j \big(\sum_{i=1}^{4} q_i \cdot y_{ij} \big) \Big] \\
\text{s.to} \quad & \sum_{i=1}^{4} r_i \cdot x_i \leq 120 \rightarrow \text{Initial investment constraint} \\
& \sum_{j=1}^{3} y_{ij} \leq x_i \quad i = 1 \ldots 4 \\
& \sum_{i=1}^{4} y_{i1} = \xi \ \ \text{(Segment one is satisfied)} \\
& \sum_{i=1}^{4} y_{ij} = d_j \qquad j = 2, 3 \\
& x_i, y_{ij} \geq 0 \qquad i = 1 \ldots 4, j = 1, \ldots, 3
\end{aligned}
$$

Stochastic Model Cont...

- Let's suppose that ξ has the following values: 3 5 7
  with probabilities: 0.3 0.4 0.3
- Solving the stochastic problem:
  - We only have to consider x_i as the 1st stage variables.
  - The y_{ij} will be the variables of the second stage, : y_{ijs} $s = 1 \ldots 3$
  - Therefore the problem is defined as:

$$
\begin{aligned}
\min \quad & \sum_{i=1}^{3} r_i \cdot x_i + \sum_{s=1}^{3} p(s) \Big[\sum_{j=1}^{3} \tau_j \big(\sum_{i=1}^{4} q_i \cdot y_{ijs} \big) \Big] \\
\text{s.to} \quad & \sum_{i=1}^{4} r_i \cdot x_i \leq 120 \\
& \sum_{j=1}^{3} y_{ijs} \leq x_i \quad i = 1 \ldots 4, s = 1 \ldots 3 \\
& \sum_{i=1}^{4} y_{i1s} = d_1(s) \ \ \text{where } d_1(s) = 3 \text{ if } s = 1; 5 \text{ if } s = 2; 7 \text{ if } s = 3 \\
& \sum_{i=1}^{4} y_{ijs} = d_j \quad j = 2, 3; s = 1, 2, 3, \text{ and } \ x_i, y_{ijs} \geq 0
\end{aligned}
$$

Stochastic Model Cont...

- Solving the last stochastic problem with AMPL:
  $x_i = (2.\widehat{6}\ \ 4\ \ 3.\widehat{3}\ \ 2) \quad Obj.F = 381.853$
- If we solved the deterministic problem (in other words, the deterministic problem obtained when we replace ξ *by* $E(\xi) = \bar{\xi}$), we get:
  $\bar{x}^*_{\bar{\xi}} = (0\ \ 3\ \ 5\ \ 2) \quad Obj.F_{\bar{\xi}} = 365$

  *** Hint: This solution is obtained by formulating the non-stochastic problem. If the stochastic problem is used fixing in AMPL that ζ takes the values 5,5,5 with probabilities 0.3, 0.4, and 0.3 we obtain the alternative optimal solution of Obj.F=365 as well, but, where $\bar{x}^*_{\bar{\xi}} = (2.\widehat{3}\ \ 0.\widehat{6}\ \ 5\ \ 2)$. Same result using MINOS.

Stochastic Model Cont...

- If we solve the stochastic problem using the expected value $\bar{x}^*_{\bar{\xi}} = (0\ \ 3\ \ 5\ \ 2)$, we obtain an infeasible problem.
- The infeasibility is due to the stochastic problem constraints that can not be satisfied when s=3 ($d_1(s) = 7$, maximum demand). In other words, if we apply this decision $x^*_{\bar{\xi}}$ with probability=0.3, we could not satisfy our demand!; this is equivalent to state that the value of the stochastic solution is:
  $VSS = +\infty - 381.85 = +\infty$
  *** Note: This never happens because the companies...:
  - they can be buy energy from external sources
  - they have technology $n + 1$...

Probabilistic or chance constraints

Probabilistic or chance constraints

- The last case of $VSS = +\infty$ can be used to introduce this concept. Let´s suppose that we have to solve the deterministic expected value problem, and we want to use this solution in the stochastic formulation, where the demand can not be satisfied. Therefore, we can add constraints as:

$$\begin{cases} \rightarrow \text{General problem formulation:} \quad P\Big(\sum_{i=1}^{n} x_i \geq \sum_{j=1}^{m} d_j(\omega)\Big) \geq \alpha \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \alpha \in [0,1] \\ \rightarrow \text{Example formulation:} \quad P\Big(\sum_{i=1}^{4} x_i \geq \sum_{j=1}^{3} d_j(\omega)\Big) \geq \alpha \\ d_j(\omega) \textit{ is} \text{ considered as a random variable.} \end{cases}$$

Probabilistic or chance constraints Cont...

- The last formulation is a probabilistic constraint used in order to guarantee certain system reliability.
  We can control the quality of the service by:

$$\downarrow$$

$$\begin{cases} \alpha \approx 1 \rightarrow \textit{Good service} \\ \alpha \approx 0 \rightarrow \textit{Bad service} \end{cases}$$

- If we know the distribution of the variable $\sum_{j=1}^{m} d_j$ (we know $F(x)$), the probabilistic constraint can be expressed as:

Probabilistic or chance constraints Cont...

$$\left\{ \begin{array}{l} \text{Example case: } P\Big(\sum_{j=1}^{m} d_j\Big) \leq \sum_{i=1}^{n} x_i\Big) \geq \alpha \equiv F_{\sum_{j=1}^{m} d_j}\Big(\sum_{i=1}^{n} x_i\Big) \geq \alpha \rightarrow \\ \rightarrow \boxed{\sum_{i=1}^{n} x_i \geq F^{-1}_{\sum_{j=1}^{m} d_j}(\alpha)} \end{array} \right.$$

Probabilistic or chance constraints Cont...

- Now these new constraints can be added in a deterministic problem as linear constraints, and we can estimate the percentile α of F.
- Adding these constraints many deterministic formulations problems are avoided in regard to stochastic formulations. Let´s see the last example:

  The r.v. is $\sum_{j=1}^{3} d_j = d_1 + d_2 + d_3 = d_1 + 3 + 2 = d_1 + 5$,

	$d_1(1)$=3+5	$d_1(2)$=5+5	$d_1(3)$=7+5
With probabilities:	0.3	0.4	0.3

Probabilistic or chance constraints Cont...

- Imposing $\alpha = 0.7 \rightarrow \sum_{i=1}^{n} x_i \geq F^{-1}_{\sum_{j=1} d_j}(0.7)$ =5+5=10
- The resulting constraint is $\sum_{i=1}^{n} x_i \geq 10$, which is satisfied by $x^*_{\bar{\xi}} = (0, 3, 5, 2)$
- If we impose $\sum_{i=1}^{n} x_i \geq F^{-1}_{\sum_{j=1} d_j}(\alpha = 1) \Leftrightarrow \sum_{i=1}^{n} x_i \geq 12$
- Therefore we get $x_{\bar{\xi},\alpha=1} = (0.8\widehat{3}, 3, 4.1\widehat{6}, 4)$, $Obj.F_{x_{\bar{\xi},\alpha=1}} = 3.78\widehat{6}$

Probabilistic or chance constraints Cont...

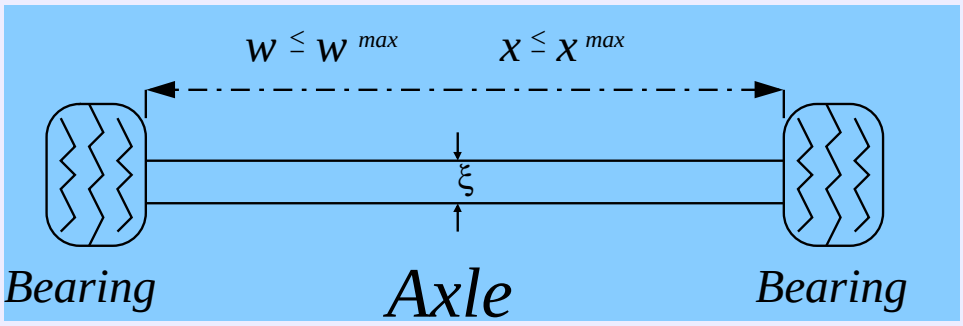
- Now, we can impose $x_{\bar{\xi},\alpha=1}$ in the stochastic problem and the solution will not be infeasible. We will obtain Obj.F=383.99, not so different from the stochastic solution: VSS=383.99-381.85=2.14
- Chance constraints allow expected value formulations to deal with stochasticity of the problem.

Design for Manufacturing Quality

- This is an example of how to manufacture a product maximizing the profit and satisfying some specific criteria of product quality.
- These types of applications of stochastic problems have been widely used in modern manufacturing (Taguchi methods).
- In fact, these methods can be seen as examples of stochastic programming, although they are often not described this way.

Design for Manufacturing Quality

- For example:
  - A manufacturing company produce axles for a bicycle cart.
  - The axles has the general appearance of:

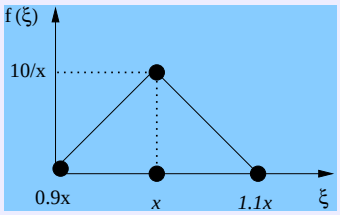


  - w (length) can be specified (deterministic)
  - The diameter ξ is a random variable that depends on a specified mean value,x, that represents, for example the setting on a machine.

Design for Manufacturing Quality Cont...

- Ideally we have that $\xi \sim N(x, \sigma)$. But, it can be simplified as:
  - ξ is a triangular random variable. Where:

$$f(\xi) = \begin{cases} \frac{100}{x^2}(\xi - 0.9x) & \text{if} \quad \xi \in [0.9x, x] \\ \frac{100}{x^2}(1.1x - \xi) & \text{if} \quad \xi \in [x, 1.1x] \\ 0 & \text{Otherwise.} \end{cases}$$



  - ξ is a uniform random variable $\xi \sim U[0.9x, 1.1x]$. This is not the most realistic case, but, it will be used to simplify the problem.

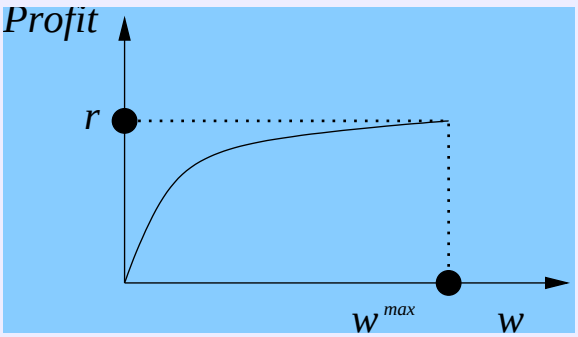
Design for Manufacturing Quality Cont...

Suppose we have 3 kinds of costs/profits in the objective function:

1. Let r be the maximum selling price per axle. The selling price for an axle with lenght w is:

$$r(1 - e^{-0.1w})$$

Supposing that $w \leq w^{max}$:



Design for Manufacturing Quality Cont...

2. The total manufacturing cost depends on the single axle material cost, therefore: $c(w\pi(\frac{\xi}{2})^2) \simeq \frac{cw\pi x^2}{4}$

3. Finally, other costs are incurred after the product is made. These costs are called quality losses. Thus, the axle quality is assured if:
   - The stress limit (for a 100 maximum central load) satisfy the next relationship:
   $$\frac{w}{\xi^3} \leq 39.27$$
   - It has a small deflection expressed as:
   $$\frac{w^3}{\xi^4} \leq 63169$$

   When some of these constraints is violated, the cost for not meeting quality is stochastic and it is assumed proportional to the square of the violation ($q \cdot y^2$), where:
   - q=cost per unit not meeting constraints.
   - y=maximum stress or deflection violation.

Design for Manufacturing Quality Cont...

- The last idea can be written as:
$$\begin{aligned} Q(w,x,\xi) = \min_{y} \quad & q \cdot y^2 \\ \text{s.to.} \quad & \frac{w}{\xi^3} - y \leq 39.27 \\ & \frac{w^3}{\xi^4} - 300 \cdot y \leq 63169 \\ & y \geq 0 \end{aligned}$$
- The expected cost, given w and x, is:
$$\mathcal{Q}(w,x) = \int_{\xi} Q(w,x,\xi) f(\xi) d\xi$$
- Our problem can thus be expressed as:
$$\begin{aligned} \max \quad & r(1 - e^{-0.1w}) - c\frac{w\pi x^2}{4} - \mathcal{Q}(x,w) \\ \text{s.to} \quad & 0 \leq w \leq w^{max} \\ & 0 \leq x \leq x^{max} \end{aligned}$$

Design for Manufacturing Quality Cont...

- The expected cost, given w, ξ and x, is:

  $Q(w,x,\xi) = q \cdot y^2$, where $y = \max(0, \frac{w}{\xi^3} - 39.27, \frac{w^3}{300\xi^4} - \frac{63169}{300})$

- Thus:
  $\mathcal{Q}(w,x) = \int_{-\infty}^{+\infty} Q(w,x,\xi) f(\xi) d\xi =$
  $= q \int_{0.9x}^{1.1x} \max(0, \frac{w}{\xi^3} - 39.27, \frac{w^3}{300\xi^4} - \frac{63169}{300}) f(\xi) d\xi$

- Even when $\xi \sim U(0.9x, 1.1x)$ the last computation is not simple.

- We can find an approximated solution when ξ is a discrete variable and:
  $w^{max} = 36$ $\quad r$=10 $\quad q$=1
  $x^{max} = 1.25$ $\quad c$=0.025
  If ξ is a discrete variable with 100 points the extended formulation is (see model on next slide):

Design for Manufacturing Quality: extended formulation

- The extended model obtained by the approximation of ξ in

$$\begin{cases} \xi_s & s = 1 \dots S \quad \text{values} \\ p_s & s = 1 \dots S \quad \text{prob's} \\ x, w & \rightarrow \ 1st \text{ stage} \\ y_s & s = 1 \dots S \ \rightarrow \text{2nd stage} \end{cases}$$

$$\begin{array}{ll} \max & r(1 - e^{-0.1w}) - c\dfrac{w\pi x^2}{4} - q\displaystyle\sum_{s=1}^{S} p_s \cdot y_s^2 \\ \text{s.to.} & \dfrac{w^3}{\xi^3} - y_s \leq 39.27 \qquad s = 1 \dots S \\ & \dfrac{w^3}{\xi^4} - 300 y_s \leq 63169 \qquad s = 1 \dots S \\ & y_s \geq 0 \qquad s = 1 \dots S \\ & 0 \leq x \leq x^{max} \quad 0 \leq w \leq w^{max} \end{array}$$

Design for Manufacturing Quality: solution

- Solution using the uniform distribution:

$w^* = 33.2681 \rightarrow$	AMPL and solve twice
$x^* = 1.0475$	for more iterations.
$Obj.F = 8.922$	

- Note: If a triangular distribution is considered the solution is:
  $x^* = 1.038$
  $w^* = 36.6$
  $Obj.F = 8.94$

VSS: the Value of the Stochastic Solution

- Solving the expected value problem (ξ replaced by $\bar{\xi} = x$):

$$
\begin{array}{ll}
\max & r(1 - e^{-0.1w}) - c\dfrac{w\pi x^2}{4} - q \cdot y^2 \\
\text{s.to.} & y \geq 0 \\
& y \geq \dfrac{w^3}{\bar{\xi}^3} - 39.27 \Leftrightarrow y \geq \dfrac{w}{x^3} - 39.27 \\
& 300y \geq \dfrac{w^3}{\bar{\xi}^4} - 63169 \Leftrightarrow 300y \geq \dfrac{w^3}{x^3} - 63169 \\
& 0 \leq x \leq x^{max} \\
& 0 \leq w \leq w^{max}
\end{array}
$$

- The solution with AMPL/MINOS:

$$
\begin{aligned}
w_{\bar{\xi}} &= 34.9702 \\
x_{\bar{\xi}} &= 0.962 \\
\text{Obj.F}_{\bar{\xi}} &= 9.0616
\end{aligned}
$$

VSS: the Value of the Stochastic Solution Cont...

- VSS $\Rightarrow$ If the solution of the expected value($w^*_{\bar{\xi}} = 34.9702$ and $x^*_{\bar{\xi}} = 0.962$) is applied on real life (stochastic problem) we get an objective function $= -131.745 < 0$!!!
  $VSS = 8.922 - (131.745) = 140.667$
- Conclusion: using the expected value solution the company has losses instead of profits by manufacturing these axles.